$$= \frac{(\sin A\cos A + \cos^2 A + \sin^2 A)(\sin A - \cos A)}{\sin A\cos A}$$

$$= \frac{(\sec A - \operatorname{cosec} A)\left(\dfrac{1}{\cos^2 A} + \dfrac{1}{\cos A \sin A} + \dfrac{1}{\sin^2 A}\right)}{(\sin A\cos A + 1)\left(\dfrac{\sin A}{\sin A\cos A} - \dfrac{\cos A}{\sin A\cos A}\right)}$$

$$= \frac{(\sin A\cos A + 1)(\sec A - \operatorname{cosec} A)}{(\sec A - \operatorname{cosec} A)\left(\dfrac{\sin^2 A + \sin A\cos A + \cos^2 A}{\sin^2 A\cos^2 A}\right)}$$

$$= \frac{(\sin A\cos A + 1)(\sec A - \operatorname{cosec} A)}{(\sec A - \operatorname{cosec} A)(1 + \sin A\cos A)} \times \sin^2 A\cos^2 A = \sin^2 A\cos^2 A$$

எடுத்துக்காட்டு 6.17 $\dfrac{\cos^2\theta}{\sin\theta} = p$ மற்றும் $\dfrac{\sin^2\theta}{\cos\theta} = q$ எனில், $p^2q^2(p^2 + q^2 + 3) = 1$ என நிரூபிக்க.

தீர்வு இங்கு $\dfrac{\cos^2\theta}{\sin\theta} = p$ ...(1) மற்றும் $\dfrac{\sin^2\theta}{\cos\theta} = q$ ...(2)

$$p^2q^2(p^2 + q^2 + 3) = \left(\frac{\cos^2\theta}{\sin\theta}\right)^2\left(\frac{\sin^2\theta}{\cos\theta}\right)^2 \times \left[\left(\frac{\cos^2\theta}{\sin\theta}\right)^2 + \left(\frac{\sin^2\theta}{\cos\theta}\right)^2 + 3\right]$$

[(1), (2) -ஐ பயன்படுத்தக் கிடைப்பது]

$$= \left(\frac{\cos^4\theta}{\sin^2\theta}\right)\left(\frac{\sin^4\theta}{\cos^2\theta}\right) \times \left[\frac{\cos^4\theta}{\sin^2\theta} + \frac{\sin^4\theta}{\cos^2\theta} + 3\right]$$

$$= (\cos^2\theta \times \sin^2\theta) \times \left[\left(\frac{\cos^6\theta + \sin^6\theta + 3\sin^2\theta\cos^2\theta}{\sin^2\theta\cos^2\theta}\right)\right]$$

$$= \cos^6\theta + \sin^6\theta + 3\sin^2\theta\cos^2\theta$$

$$= (\cos^2\theta)^3 + (\sin^2\theta)^3 + 3\sin^2\theta\cos^2\theta$$

$$= [(\cos^2\theta + \sin^2\theta)^3 - 3\cos^2\theta\sin^2\theta(\cos^2\theta + \sin^2\theta)] + 3\sin^2\theta\cos^2\theta$$

$$= 1 - 3\cos^2\theta\sin^2\theta(1) + 3\cos^2\theta\sin^2\theta = 1$$

முன்னேற்றச் சோதனை

1. முக்கோணவியல் விகிதங்களின் எண்ணிக்கையானது _____.
2. $1 - \cos^2\theta$ = _____.
3. $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta)$ = _____.
4. $(\cot\theta + \operatorname{cosec}\theta)(\cot\theta - \operatorname{cosec}\theta)$ = _____.
5. $\cos 60^\circ \sin 30^\circ + \cos 30^\circ \sin 60^\circ$ = _____.
6. $\tan 60^\circ \cos 60^\circ + \cot 60^\circ \sin 60^\circ$ = _____.
7. $(\tan 45^\circ + \cot 45^\circ) + (\sec 45^\circ \operatorname{cosec} 45^\circ)$ = _____.
8. (i) $\sec\theta = \operatorname{cosec}\theta$ எனில், θ = _____. (ii) $\cot\theta = \tan\theta$ எனில், θ = _____.

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